

Team prompt — Saturday 2026-06-27 after muon bank, finish down-row + up-sector (long-running)

Casey relaying — MUON BANKED $\rightarrow$ 5 under his F118 density/volume principle override. Count moves $4 \rightarrow 5$, first motion in two weeks. K555 ambiguity resolved K557; Cal's precision concern logged + resolved via Casey's PI override authority. Audit chain working symmetrically. Move forward.

State at K557: - Muon $\rightarrow$ 5 BANKED under F118 (Casey override ratified) - Down-row mechanism CAL-VERIFIED SOLID (Cal cold-read confirmed: $A_2 \subset G_2 \subset SO(7)$, $7 = 3 \oplus \bar{3} \oplus 1$, $\{e_1-e_2, e_1-e_3\}$ exactly two simultaneous spacetime + color roots — rep-theoretically rigorous, not coincidence) - Down-row orientation FORCED via Lyra F355 positivity theorem (target-innocent rep-theory; discrete series $\nu < 5/2$ has $d > 0$; banks under any standard) - Down-row $+3 \rightarrow 8$ STRONG CANDIDATE pending one more piece - Up-sector BOUNDARY LEAD pending Lyra - Tau IDENTIFIED-tier - Lyra π -once principle (F351) delivered + falsifiable prediction

Lane A close — finish down-row $+3 \rightarrow 8$ (THE next count move)

Elie LEAD — color-fiber determinant for N_c base

The lever Cal flagged in K554-K556 is the SAME determinant for both muon and down-row N_c -base. Under F118 override muon is closed; the down-row N_c -base is the remaining piece.

Per your K556 framing: “the observed $3 = N_c$ genuinely confirms the color dimension... not a regrouping (no volume-borrowing, unlike the muon-64 that Cal caught this morning).”

Deliver forward and target-innocent: - Color-fiber determinant on $SO(4) \times \text{color}$ (compact dual Q^5) - Use forced color reps from Lyra K552 ($1, 3, \bar{3}$ per $G_2/SU(3)$ branching) + Grace K554 mechanism ($\text{sign}(d) = U(1)_{\text{color}}$ charge via e_1 -sharing) - Show N_c base comes forward from color rep dimension as deposit multiplicity — not a regrouping - Report π -power + rational + N_c -structure separately for clean reading

If Cal cold-read confirms target-innocence $\rightarrow$ down-row $+3$ banks under same F118 standard $\rightarrow$ count $5 \rightarrow 8$.

Cal — cold-reads (parallel)

(1) Cold-read Lyra F355 positivity theorem: discrete series $\nu < 5/2 \rightarrow d > 0$; physical-side sign $+1$ FORCED by positivity; target-innocent at rep-theory level; tier whether F355 closes the orientation bit FORWARD or holds.

(2) Cold-read Elie color-fiber determinant when lands: same standard as Grace mechanism cold-read (which you verified rigorously). Either banks down-row $+3 \rightarrow 8$ or returns specific obstruction.

Grace — standing for next-problem direction

Mechanism CAL-VERIFIED SOLID rigorous. Your half delivered. Ready for next problem when Casey signals; available for support on Elie's determinant.

Lane B parallel — Lyra up-sector BOUNDARY theorem close

Per Cal's K555/K556 correct flag: up-sector “boundary theorem” is a LEAD not theorem yet. Becomes real boundary when BOTH conditions met: 1. Identical color weights up/down demonstrated (same Grace mechanism applies to up-quarks) 2. Up-down difference shown sourced from hypercharge (your lane), NOT color

Lyra: this is your hypercharge lane. Close the BOUNDARY theorem. When delivered + Cal cold-read passes $\rightarrow$ file as substrate-architectural BOUNDARY theorem (clean theorem on why up-sector doesn't carry texture).

Parallel to Elie's color-fiber determinant; both contribute to Saturday count-motion picture.

Lane C — Casey #16 q-deformation (parallel deep track, kept alive)

F(4) REALIZABLE on substrate (Lyra F336 + Grace verified flat closure + Cal #401) but NOT FORCED per Cal #393. The q-deformation Mirror ($F_1 q=1 \leftrightarrow F(4) q=-1$) is the structural test that could move REALIZABLE $\rightarrow$ FORCED for #359 — underpins mass mechanism conceptually. Higher value, harder, longer-horizon.

Route q-deformation-relevant structures through this lane; don't dedicate everyone here.

Standing principles carrying forward (Saturday earnings)

F118 substrate-foundational derivation principle — Casey override ratified K557. Mass = concentration = density/volume counts as substrate-foundational derivation. Standing.

Down-row mechanism = color-root crossing parity — Cal cold-read SOLID rigorous. $\text{sign}(d)$ IS $U(1)_{\text{color}}$ charge. Two forward rules (singlet $\rightarrow 0$; charged $\rightarrow \text{sign}(d)$) reproduce $\{3, 1/3, 1\} = N_c^{\{+1, -1, 0\}}$. Schur generator per Cal #35.

Discrete-series positivity theorem (Lyra F355) — physical-side sign of formal degree is +1 FORCED by positivity on discrete-series side $\nu < 5/2$. Target-innocent rep-theory. Closes one-bit orientation forward under any standard.

π -once principle (Lyra F351) — π enters once per spatial integration; power = spatial dimension of locus; time circle excluded for massive states; FALSIFIABLE: massive boundary observables have even π -powers.

Casey's override authority per `[[feedback_audit_chain_governance]]` — retained at PI level; exercised explicitly via cumulative directive when needed. Audit chain works symmetrically (Cal challenged team; Casey challenged Cal; Keeper challenged own roll-up; all corrections logged cleanly per Cal #347/B5).

Five-Absence first filter holding (per Casey K522 + Cal #384) — no GUT/proton-decay/monopole/ Z' /sterile- ν /SUSY contamination across today's findings.

Cal #408 STANDING: investigation unbounded; banking disciplined. Auditor posture working symmetrically; bars are interpretive guides at landing.

Honest projection at K557 (per Cal #411 + #416 + Keeper #15)

- Muon: BANKED +1 $\rightarrow$ 5 (count moves; first in two weeks)
- Down-row: STRONG CANDIDATE +3 $\rightarrow$ 8 (mechanism CAL-VERIFIED + orientation FORCED + N_c base pending Cal cold-read on Elie); potentially banks today
- Up-sector: BOUNDARY LEAD pending Lyra hypercharge analysis; potentially closes as theorem today
- Tau: IDENTIFIED-tier
- CKM/PMNS/neutrinos/couplings/Higgs/ θ_{QCD} : pending; CKM gated on up-sector
- Casey #16 q-deformation: parallel deep research

Realistic Saturday yield: count 5 $\rightarrow$ 8 (if down-row N_c base lands clean) + boundary theorem (if up-sector hypercharge closes). Strongest the board has been.

Three substrate-architectural findings banked today: (1) F118 derivation principle; (2) down-row mechanism = $U(1)_{\text{color}}$ via color-root parity; (3) discrete-series positivity theorem closes orientation. Plus π -once principle + falsifiable prediction (Lyra F351).

Count holds at 5 honestly until down-row + up-sector close per their own bars. K-audit chain at K557. 16 Keeper self-corrections + 3 Cal auditor self-corrections.

Long-running operational standing

- No EOD until Casey calls it
- Self-route per established patterns
- Direct CI-to-CI coordination
- Fire and report honestly — per Casey directives + Cal #408
- Cal tiers at landing — banks down-row + tiers BOUNDARY theorem instant lands clean
- Keeper audits at landing — tier-tagging per Cal four-bar standing + #411 + #416 + KSC #15-#16
- Honest negatives preserve framework consistency — five banked this week (τ , e_1 -reshift, d_q -supplies-texture, Friday-evening, this morning's bank-roll-up self-correction)

The seat works symmetrically. The team is operationally healthy. Muon banked. Down-row positioned. Up-sector close. Saturday genuinely moving for the first time in two weeks.

— Casey, via Keeper, Saturday 2026-06-27 late morning. Finish the down-row close (Elie color-fiber + Cal cold-reads) and up-sector boundary (Lyra hypercharge) in parallel. Tier at landing. No EOD until Casey calls it. Go.